1 Multiplication: convolution of logarithmic blocks

Multiplication is the operation on which every later construction rests: reciprocals, powers, composition, and reversion are all built from it by triangular recurrences. Its whole content is that the Cauchy rule in the outer variable t is *coefficientwise finite*, so that each output block is an exact finite expression in finitely many input blocks. Nothing here is an analytic rearrangement, and nothing here requires convergence.

Throughout this section $\mathbb{K}$ is a field of characteristic zero, X is the large variable, the regime is $X \to +\infty$ on the positive ray unless a different domain is announced, and

$$t := X^{-1}, \qquad L := \log X. \tag{1}$$

In the formal algebra t and L are independent generators; their analytic coupling is recorded only by the distinguished derivation, which plays no role in this section. The four ambient classes are

class	coefficient ring	elements
$\mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}}$	$\mathbb{K}[L]$	$\sum_{n\geq 0} p_n(L)t^n$
$\mathscr{P}\mathscr{L}_{\mathrm{poly},\mathbb{K}}$	$\mathbb{K}[L]$	$\sum_{n\geq n_0} p_n(L)t^n$, $n_0 \in \mathbb{Z}$
$\mathscr{P}\mathscr{L}_{\mathrm{rat},\mathbb{K}}$	$\mathbb{K}(L)$	$\sum_{n\geq n_0} p_n(L)t^n$, p_n rational in L
$\mathscr{P}\mathscr{L}_{\mathrm{itLaur},\mathbb{K}}$	$\mathbb{K}((L^{-1}))$	$\sum_{n\geq n_0} t^n \sum_{j\geq j_n} a_{n,j}L^{-j}$

All truncations below are *exclusive*: $\mathrm{Trunc}_{<N} F$ keeps every logarithmic monomial in every block t^n with $n < N$, and discards the rest.

Remark 1.1 (Reconciliation of the source conventions). Two of the merged reports use an inclusive outer cutoff $[F]_{\leq N}$. The translation is $[F]_{\leq N} = \mathrm{Trunc}_{<N+1} F$, and every dependency range, algorithm bound and error exponent quoted from those reports has been shifted by one block accordingly. Likewise, one report writes the coefficient bracket as $[F]_M$; here the series always stands to the right of the bracket, $[t^n]F = p_n(L)$ and $[t^n L^j]F = a_{n,j}$.

1.1 The block Cauchy product

It is economical to prove the algebra of all four classes at once. What the three coefficient rings $\mathbb{K}[L]$, $\mathbb{K}(L)$, $\mathbb{K}((L^{-1}))$ have in common is a degree function in L together with a coefficient-extraction map obeying the usual convolution rule.

Definition 1.2 (Logarithmically graded domain)**.** A *logarithmically graded domain over* $\mathbb{K}$ is a commutative $\mathbb{K}$-algebra R together with $\mathbb{K}$-linear maps $\kappa_d \colon R \to \mathbb{K}$, one for each $d \in \mathbb{Z}$, such that for every nonzero $f \in R$ the set $\{d : \kappa_d(f) \neq 0\}$ is nonempty and bounded above, and:

(G1) $f = 0$ if and only if $\kappa_d(f) = 0$ for every d;

(G2) $\kappa_d(fg) = \sum_{i+j=d} \kappa_i(f)\kappa_j(g)$ for all $f, g \in R$ and all $d \in \mathbb{Z}$, the sum having only finitely many nonzero terms;

(G3) $\kappa_0(1) = 1$ and $\kappa_d(1) = 0$ for $d \neq 0$.

For $f \neq 0$ put $\deg_{\mathrm{L}} f := \max\{d : \kappa_d(f) \neq 0\}$ and $\mathrm{lc}_L(f) := \kappa_{\deg_{\mathrm{L}} f}(f) \in \mathbb{K}^\times$; set $\deg_{\mathrm{L}} 0 := -\infty$ and $\mathrm{lc}_L(0) := 0$.

Here lc_L is the leading coefficient *inside one block*; it is not the same object as the leading scalar lc of a series, defined in Theorem 1.18 below, although the two agree on the leading block.

Lemma 1.3 (Elementary consequences)**.** *Let R be a logarithmically graded domain over $\mathbb{K}$. For all $f, g \in R$:*

$$\deg_{\mathrm{L}}(fg) = \deg_{\mathrm{L}} f + \deg_{\mathrm{L}} g, \qquad \mathrm{lc}_L(fg) = \mathrm{lc}_L(f)\,\mathrm{lc}_L(g), \qquad \deg_{\mathrm{L}}(f+g) \leq \max(\deg_{\mathrm{L}} f, \deg_{\mathrm{L}} g). \tag{2}$$

Here $-\infty + c = -\infty$ *and* $\max(-\infty, c) = c$*. In particular* R *is an integral domain. Moreover, if* $\deg_{\mathrm{L}} f \le d$ *and* $\deg_{\mathrm{L}} g \le e$*, then* $\kappa_{d+e}(fg) = \kappa_d(f)\kappa_e(g)$*.*

Proof. If $f = 0$ or $g = 0$ every assertion is trivial under the stated conventions. The third assertion in (2) is immediate from $\mathbb{K}$-linearity of each κ_d. For the first two, assume $f, g \neq 0$ and set $d = \deg_{\mathrm{L}} f$, $e = \deg_{\mathrm{L}} g$. If $c > d + e$ and $i + j = c$, then $i > d$ or $j > e$, so every term of (G2) vanishes and $\kappa_c(fg) = 0$. If $c = d + e$, the only possibly nonvanishing term is $i = d$, $j = e$, whence

$$\kappa_{d+e}(fg) = \kappa_d(f)\kappa_e(g) = \mathrm{lc}_L(f)\,\mathrm{lc}_L(g) \neq 0,$$

because $\mathbb{K}$ is a field and both factors are nonzero. This proves $\deg_{\mathrm{L}}(fg) = d + e$, $\mathrm{lc}_L(fg) = \mathrm{lc}_L(f)\,\mathrm{lc}_L(g)$, and, by (G1), $fg \neq 0$; so R is a domain. The last sentence is the same computation with d, e merely upper bounds, all other terms of (G2) again vanishing. □

Lemma 1.4 (The three coefficient rings)**.** *Each of*

$$\mathbb{K}[L], \qquad \mathbb{K}(L), \qquad \mathbb{K}((L^{-1}))$$

is a logarithmically graded domain over $\mathbb{K}$*, with* $\deg_{\mathrm{L}}$ *equal respectively to the polynomial degree, to* $\deg_{\mathrm{L}}(p/q) = \deg p - \deg q$*, and to the largest exponent occurring in the Laurent expansion. There are injective* $\mathbb{K}$*-algebra homomorphisms*

$$\mathbb{K}[L] \hookrightarrow \mathbb{K}(L) \hookrightarrow \mathbb{K}((L^{-1})) \tag{3}$$

that preserve $\deg_{\mathrm{L}}$ *and* lc_L*.*

Proof. For $\mathbb{K}[L]$ take $\kappa_d(f)$ to be the coefficient of L^d in f for $d \ge 0$ and $\kappa_d = 0$ for $d < 0$; (G1)–(G3) are the definition of polynomial multiplication. For $\mathbb{K}((L^{-1}))$, an element is $f = \sum_{j \le d} a_j L^j$ with only finitely many $j > 0$ allowed but arbitrarily negative j permitted; put $\kappa_j(f) = a_j$. Then (G1) and (G3) are clear, and in (G2) the constraint $i + j = c$ with $i \le \deg_{\mathrm{L}} f$, $j \le \deg_{\mathrm{L}} g$ forces $i \ge c - \deg_{\mathrm{L}} g$, so the sum is finite; it is the definition of the Laurent product. The support of f is bounded above by construction.

The inclusion $\mathbb{K}[L] \subset \mathbb{K}((L^{-1}))$ is a $\mathbb{K}$-algebra map, and every nonzero element of $\mathbb{K}((L^{-1}))$ is invertible: writing $f \neq 0$ as $f = a_d L^d(1 + u)$ with $d = \deg_{\mathrm{L}} f$ and $u \in L^{-1}\mathbb{K}[[L^{-1}]]$, the family $\big((-u)^r\big)_{r \ge 0}$ is L^{-1}-adically summable and its sum inverts $1 + u$, so $f^{-1} = a_d^{-1} L^{-d} \sum_{r \ge 0} (-u)^r$. Hence every nonzero polynomial becomes invertible, and the universal property of the field of fractions supplies a unique $\mathbb{K}$-algebra homomorphism $\iota\colon \mathbb{K}(L) \to \mathbb{K}((L^{-1}))$ extending the inclusion; it is injective because $\mathbb{K}(L)$ is a field and $\iota(1) = 1$. Pulling the maps κ_d back along ι makes $\mathbb{K}(L)$ a logarithmically graded domain, and Theorem 1.3 applied to $\iota(p) = \iota(q)\iota(p/q)$ gives $\deg_{\mathrm{L}}(p/q) = \deg p - \deg q$ and $\mathrm{lc}_L(p/q) = \mathrm{lc}_L(p)\,\mathrm{lc}_L(q)^{-1}$. Both maps in (3) preserve κ_d by construction, hence preserve $\deg_{\mathrm{L}}$ and lc_L. □

Definition 1.5 (The block Cauchy product)**.** Let R be a logarithmically graded domain over $\mathbb{K}$ and let

$$F = \sum_{n \ge a} p_n\, t^n, \qquad G = \sum_{m \ge b} q_m\, t^m, \qquad p_n, q_m \in R, \tag{4}$$

be elements of $R((t))$, the ring of formal Laurent series in t over R. Their product is

$$FG := \sum_{N \ge a+b} C_N\, t^N, \qquad C_N := \sum_{\substack{n+m=N \\ n \ge a,\ m \ge b}} p_n q_m = \sum_{n=a}^{N-b} p_n\, q_{N-n}. \tag{5}$$

Lemma 1.6 (Finiteness of every fiber)**.** *For fixed* N *the index set in* (5) *is* $\{n \in \mathbb{Z} : a \le n \le N - b\}$*, of cardinality* $\max(N - a - b + 1, 0)$*. In particular it is finite, and it is empty exactly when* $N < a + b$*.*

Proof. The constraints $n \geq a$ and $m = N - n \geq b$ are equivalent to $a \leq n \leq N - b$. An interval of integers $[a, N - b]$ has $N - b - a + 1$ elements when $N - b \geq a$ and is empty otherwise. □

This is the entire mechanism. There is no limit, no rearrangement, and no hypothesis of numerical convergence: each C_N is a finite sum of products in R.

Theorem 1.7 (Closure of the four ambient classes). *Let R be a logarithmically graded domain over $\mathbb{K}$. Then (5) makes $R((t))$ a commutative $\mathbb{K}$-algebra with unit 1, containing $R[[t]]$ as a subalgebra. Consequently each of*

$$\mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}} = \mathbb{K}[L][[t]], \quad \mathscr{P}\mathscr{L}_{\mathrm{poly},\mathbb{K}} = \mathbb{K}[L]((t)), \quad \mathscr{P}\mathscr{L}_{\mathrm{rat},\mathbb{K}} = \mathbb{K}(L)((t)), \quad \mathscr{P}\mathscr{L}_{\mathrm{itLaur},\mathbb{K}} = \mathbb{K}((L^{-1}))((t))$$

is closed under multiplication, and the chain of inclusions

$$\mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}} \subset \mathscr{P}\mathscr{L}_{\mathrm{poly},\mathbb{K}} \subset \mathscr{P}\mathscr{L}_{\mathrm{rat},\mathbb{K}} \subset \mathscr{P}\mathscr{L}_{\mathrm{itLaur},\mathbb{K}} \tag{6}$$

consists of injective $\mathbb{K}$-algebra homomorphisms.

Proof. By Theorem 1.6 each C_N is a well-defined element of R, and $C_N = 0$ for $N < a + b$; so FG again has support bounded below and lies in $R((t))$. Commutativity is the symmetry of the index set under $(n, m) \mapsto (m, n)$; bilinearity over $\mathbb{K}$ is clear; and $1 \cdot F = F$ because the only admissible pair for $1 = 1 \cdot t^0$ at level N is $(0, N)$. For associativity let $H = \sum_{k \geq c} h_k t^k$. Then

$$[t^N]\big((FG)H\big) = \sum_{r+k=N} \Big(\sum_{n+m=r} p_n q_m \Big) h_k = \sum_{n+m+k=N} p_n q_m h_k,$$

the last sum ranging over the finite set of triples with $n \geq a$, $m \geq b$, $k \geq c$: indeed $n + m + k = N$ with the three lower bounds confines each index to a finite interval, so the regrouping is a finite rearrangement. The same computation with the other bracketing gives the same finite sum, whence $(FG)H = F(GH)$. Distributivity is likewise a finite regrouping.

If $\nu_t F \geq 0$ and $\nu_t G \geq 0$, then $C_N = 0$ for $N < 0$, so $R[[t]]$ is closed under the product; it is a subalgebra. Taking $R = \mathbb{K}[L]$ gives the first two classes, $R = \mathbb{K}(L)$ the third, $R = \mathbb{K}((L^{-1}))$ the fourth, using Theorem 1.4. Finally, a $\mathbb{K}$-algebra homomorphism $\varphi\colon R \to R'$ induces a $\mathbb{K}$-algebra homomorphism $R((t)) \to R'((t))$ by $\sum p_n t^n \mapsto \sum \varphi(p_n) t^n$, because φ commutes with the finite sums (5); injectivity is inherited blockwise. Applying this to (3) yields (6). □

Proposition 1.8 (The monomial-level convolution and the support bound). *Let $F = \sum_{n,j} a_{n,j} t^n L^j$ and $G = \sum_{m,k} b_{m,k} t^m L^k$ lie in $\mathscr{P}\mathscr{L}_{\mathrm{poly},\mathbb{K}}$ (so $j, k \in \mathbb{N}_0$, and $\mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}}$ is the case $\nu_t \geq 0$) or in $\mathscr{P}\mathscr{L}_{\mathrm{itLaur},\mathbb{K}}$ (so $j, k \in \mathbb{Z}$, bounded above at each fixed outer level). Then*

$$[t^N L^d](FG) = \sum_{n+m=N} \sum_{i+j=d} a_{n,i} b_{m,j}, \tag{7}$$

a finite sum, and

$$\mathrm{supp}_{\mathrm{coeff}}(FG) \subseteq \mathrm{supp}_{\mathrm{coeff}}(F) + \mathrm{supp}_{\mathrm{coeff}}(G)\,, \tag{8}$$

the Minkowski sum in $\mathbb{Z}^2$.

Proof. Formula (7) is (G2) applied to (5). Finiteness holds because the outer sum is finite by Theorem 1.6 and, for each of its terms, the inner sum is finite by (G2). For (8): if $(N, d) \notin \mathrm{supp}_{\mathrm{coeff}}(F) + \mathrm{supp}_{\mathrm{coeff}}(G)$, then every summand of (7) has $a_{n,i} = 0$ or $b_{m,j} = 0$, so $[t^N L^d](FG) = 0$. □

Remark 1.9 ($\mathscr{P}\mathscr{L}_{\mathrm{rat},\mathbb{K}}$ has no monomial support of its own). Theorem 1.8 is deliberately not stated for $\mathscr{P}\mathscr{L}_{\mathrm{rat},\mathbb{K}}$. An element of $\mathbb{K}(L)((t))$ has coefficient blocks that are rational functions of L, and a rational function is not a (finite or infinite) combination of powers of L until one chooses an expansion point. The coefficient support $\mathrm{supp}_{\mathrm{coeff}}(\cdot)$ becomes defined only after applying the embedding $\mathscr{P}\mathscr{L}_{\mathrm{rat},\mathbb{K}} \hookrightarrow \mathscr{P}\mathscr{L}_{\mathrm{itLaur},\mathbb{K}}$ of Theorem 1.7, which expands each block at $L = \infty$ and turns the exact coefficient $1/(L + 1)$ into the infinite series $L^{-1} - L^{-2} + L^{-3} - \cdots$. Exact rational dependence on L and a monomial support in L are different pieces of data; only the second is what (8) constrains.

Remark 1.10 (Why multiplication is a two-level, not a flat, operation). Although (7) is a genuine two-dimensional convolution, it should not be implemented as one. The outer index is the asymptotic hierarchy and the inner index is not: at fixed N the inner sum is bounded, whereas a cutoff imposed jointly on $|N| + |d|$ does not respect dominance. See Theorem 1.17.

Remark 1.11 (Hahn supports). Nothing in Theorem 1.6 uses that the exponents are integers, only that the two supports are well ordered in the direction of increasing smallness. If $S, T \subseteq \Gamma$ are reverse-well-ordered subsets of an ordered abelian group, then $S + T$ is reverse-well-ordered and each $\gamma \in S + T$ has only finitely many decompositions $\gamma = s + t$ with $s \in S$, $t \in T$; this is the Hahn support theorem [?, ?]. Everything in this section that does not mention the specific value group $\mathbb{Z}$ transfers verbatim to the Hahn power-log extension $\sum_{\alpha \in S} p_\alpha(L) t^\alpha$ and, in particular, to the Puiseux-log extensions $\mathbb{K}[L]((t^{1/s}))$ used later for fractional powers. We do not repeat the statements.

1.2 Finite determination and precision bookkeeping

The content of Theorem 1.6 deserves to be recorded as a dependency statement, because that is the form in which it is used: it says exactly how much of the input must be known in order to know a given part of the output, and it says that no more is needed and no less will do.

Theorem 1.12 (Finite determination, with exact dependency range)**.** *Let R be a logarithmically graded domain over $\mathbb{K}$, let $a, b, N \in \mathbb{Z}$, and let $F, G \in R((t))$ satisfy $\nu_t F \geq a$ and $\nu_t G \geq b$. Then*

$$[t^N](FG) = \sum_{n=a}^{N-b} p_n \, q_{N-n}, \tag{9}$$

so that $[t^N](FG)$ is a $\mathbb{K}$-bilinear function of the two finite tuples

$$(p_a, p_{a+1}, \ldots, p_{N-b}) \qquad \text{and} \qquad (q_b, q_{b+1}, \ldots, q_{N-a}),$$

equivalently of $\operatorname{Trunc}_{<N-b+1} F$ and $\operatorname{Trunc}_{<N-a+1} G$. Both ranges are exact: for every n_0 with $a \leq n_0 \leq N - b$ there are $F, F' \in R((t))$ with $\nu_t \geq a$ that differ only in the block of index n_0, and a G with $\nu_t G \geq b$, such that $[t^N](FG) \neq [t^N](F'G)$; and symmetrically in the second argument.

Proof. Equation (9) and the bilinearity are Theorem 1.5 together with Theorem 1.6; the blocks listed are precisely those occurring in the sum. For exactness fix $n_0 \in [a, N-b]$ and put $m_0 = N - n_0$, so that $m_0 \geq b$. Take $G = t^{m_0}$, which has $\nu_t G = m_0 \geq b$. If $n_0 > a$ set $F = t^a + t^{n_0}$ and $F' = t^a$; if $n_0 = a$ set $F = t^a$ and $F' = 2t^a$ (here $2 \neq 0$ because $\operatorname{char} \mathbb{K} = 0$). In both cases $\nu_t F = \nu_t F' = a$, the two series agree outside index n_0, and $[t^N](FG) - [t^N](F'G)$ equals 1 respectively -1. The symmetric statement follows by commutativity. $\square$

Structural checkpoint

The precision rule. If F is known through $\operatorname{Trunc}_{<P} F$ and G through $\operatorname{Trunc}_{<Q} G$, and $\nu_t F = a$, $\nu_t G = b$, then FG is known through

$$\operatorname{Trunc}_{<\min(P+b,\, Q+a)} FG$$

and, in the worst case, no further. The binding entry is contributed by whichever factor is paired with the *more singular* partner: a factor known to P blocks is degraded by the valuation b of the other factor. In particular, multiplying by a series with a pole of order r, that is with valuation $-r$, costs exactly r blocks of precision.

Corollary 1.13 (Truncated product)**.** *Let $F, G \in R((t))$ be nonzero with $a = \nu_t F$, $b = \nu_t G$, and let $N \in \mathbb{Z}$. Then for all integers $P \geq N - b$ and $Q \geq N - a$,*

$$\operatorname{Trunc}_{<N} FG = \operatorname{Trunc}_{<N}\big(\operatorname{Trunc}_{<P} F\big)\big(\operatorname{Trunc}_{<Q} G\big); \tag{10}$$

and if $N > a+b$ the pair $(P,Q) = (N-b,\ N-a)$ is optimal: neither entry can be lowered by one, that is, there exist F, G with these valuations for which $P = N-b-1$, respectively $Q = N-a-1$, falsifies (10). If $F, G \in \mathscr{PL}_{\mathrm{pow},\mathbb{K}}$ and $N \geq 0$, then $(P,Q) = (N,N)$ is admissible, so the cruder law

$$\operatorname{Trunc}_{<N} FG = \operatorname{Trunc}_{<N}\big(\operatorname{Trunc}_{<N} F\big)\big(\operatorname{Trunc}_{<N} G\big) \qquad (N \geq 0) \tag{11}$$

holds there.

Proof. Write $F = \operatorname{Trunc}_{<P} F + E$ and $G = \operatorname{Trunc}_{<Q} G + H$, so that $\nu_t E \geq P$ and $\nu_t H \geq Q$, while $\nu_t \operatorname{Trunc}_{<P} F \geq a$ and $\nu_t \operatorname{Trunc}_{<Q} G \geq b$. Then

$$FG - \big(\operatorname{Trunc}_{<P} F\big)\big(\operatorname{Trunc}_{<Q} G\big) = E\ \operatorname{Trunc}_{<Q} G + \operatorname{Trunc}_{<P} F\ H + EH,$$

whose ν_t is at least $P+b \geq N$, $a+Q \geq N$ and $P+b \geq N$ respectively; for the third term note that $H = G - \operatorname{Trunc}_{<Q} G$ has $\nu_t H \geq b$, every block of H being a block of G, so $\nu_t(EH) \geq P+b$. Hence the two products agree below t^N, which is (10).

For the optimality and the sharpness, assume $N > a+b$, so that $N-1 \geq a+b$. By Theorem 1.12 the block $[t^{N-1}](FG)$ already depends on p_{N-1-b} and on q_{N-1-a}, so no pair below $(N-b, N-a)$ can serve for all F, G; the following pair of series shows that neither entry may be lowered by one. If $N-b-1 > a$ put $F = t^a + t^{N-b-1}$; if $N-b-1 = a$ put $F = t^a$. In both cases take $G = t^b$. Then $[t^{N-1}](FG) = 1$, whereas $\operatorname{Trunc}_{<N-b-1} F$ omits the block t^{N-b-1}, so $\operatorname{Trunc}_{<N}(\operatorname{Trunc}_{<N-b-1} F)G$ has zero block at t^{N-1}. (In the first case $N-1 > a+b$, so the surviving term t^{a+b} does not sit at level $N-1$.) The second cutoff is symmetric.

For (11), note that $a, b \geq 0$ gives $N \geq N-b$ and $N \geq N-a$, so $(P,Q) = (N,N)$ is admissible. ☐

A common trap

The crude law (11) is *false* on the Laurent ring. Take $F = t^{-1}$, $G = 1+t$, $N = 1$. Then $FG = t^{-1} + 1$, so $\operatorname{Trunc}_{<1} FG = t^{-1}+1$, whereas $\operatorname{Trunc}_{<1} F \operatorname{Trunc}_{<1} G = t^{-1} \cdot 1 = t^{-1}$. The correct cutoff for G is $N - \nu_t F = 2$, and indeed $t^{-1}(1+t) = t^{-1}+1$. Implementations that "normalize every series to start at t^0" silently commit this error whenever a genuine principal part is present.

Corollary 1.14 (Error rule for approximate factors). *Let $F, G \in \mathscr{PL}_{\mathrm{poly},\mathbb{K}}$ with $\nu_t F = a$, $\nu_t G = b$, and suppose $\widetilde{F}, \widetilde{G} \in \mathscr{PL}_{\mathrm{poly},\mathbb{K}}$ satisfy*

$$F = \widetilde{F} + \mathcal{O}_t^{\mathrm{form}}(t^p)\,, \qquad G = \widetilde{G} + \mathcal{O}_t^{\mathrm{form}}(t^q)\,.$$

Then

$$FG = \widetilde{F}\widetilde{G} + \mathcal{O}_t^{\mathrm{form}}\Big(t^{\min(p+b,\ q+a,\ p+q)}\Big)\,. \tag{12}$$

If in addition $p \geq a$ and $q \geq b$ — that is, if neither approximation is coarser than the leading order of the quantity it approximates — then the third entry is redundant and

$$FG = \widetilde{F}\widetilde{G} + \mathcal{O}_t^{\mathrm{form}}\Big(t^{\min(p+b,\ q+a)}\Big)\,. \tag{13}$$

Proof. Write $E = F - \widetilde{F}$ and $H = G - \widetilde{G}$, so $\nu_t E \geq p$ and $\nu_t H \geq q$. Expanding,

$$FG - \widetilde{F}\widetilde{G} = E\widetilde{G} + \widetilde{F}H + EH. \tag{14}$$

Now $\widetilde{G} = G - H$ gives $\nu_t \widetilde{G} \geq \min(\nu_t G, \nu_t H) \geq \min(b,q)$, and likewise $\nu_t \widetilde{F} \geq \min(a,p)$. Since ν_t is additive on products (Theorem 1.21 below, or directly Theorem 1.6) and satisfies $\nu_t(A+B) \geq \min(\nu_t A, \nu_t B)$,

$$\nu_t\big(FG - \widetilde{F}\widetilde{G}\big)\ \geq\ \min\big(p + \min(b,q),\ \min(a,p) + q,\ p+q\big)\ =\ \min(p+b,\ q+a,\ p+q),$$

which is (12). If $p \geq a$ and $q \geq b$, then $p+q \geq a+q$ and $p+q \geq p+b$, so $p+q \geq \min(p+b, q+a)$ and the third entry may be dropped. ☐

Remark 1.15 (What (12) does and does not say). The symbol $\mathcal{O}_t^{\mathrm{form}}(t^N)$ asserts coefficient agreement below outer order N and nothing else: no numerical bound, no convergence, no uniformity in L. It is also not a statement about a quotient ring. In $\mathscr{PL}_{\mathrm{poly},\mathbb{K}}$ the element t is a unit, so the ideal generated by t^N is the whole ring; the subset $t^N\mathscr{PL}_{\mathrm{pow},\mathbb{K}}$ is an additive piece of the valuation filtration, not an ideal. Congruences of the form (12) must therefore be read as valuation inequalities $\nu_t(F-G)\geq N$, or equivalently as equalities of exclusive truncations. Quotient-algebra language becomes legitimate only after the finite principal part has been removed and one works inside $\mathscr{PL}_{\mathrm{pow},\mathbb{K}}$, where (t^N) is a genuine ideal.

The corresponding algorithm is the schoolbook convolution restricted to the window supplied by Theorem 1.12.

Listing 1: Truncated block product

```
# F, G sparse maps  exponent -> coefficient in R;  a = ord(F), b = ord(G)
# returns the blocks of Trunc_{<N}(F*G)
function block_product(F, G, a, b, N):
    C := empty map
    for (n, Pn) in F with a <= n <= N-b-1:
        for (m, Qm) in G with b <= m <= N-n-1:
            C[n+m] := C.get(n+m, 0) + Pn*Qm
    normalize every coefficient of C          # collect in L before zero-testing
    delete the blocks of C that are exactly 0
    return C
```

Remark 1.16 (Cost, and what actually dominates it). With dense consecutive support the loop performs $\mathcal{O}\big((N-a-b)^2\big)$ coefficient products; with sparse supports of sizes s_F, s_G it performs at most $s_F s_G$. If the retained blocks have logarithmic degree $\mathcal{O}(d)$, each coefficient product costs $\mathcal{O}(d^2)$ scalar operations by schoolbook polynomial arithmetic, so the total is $\mathcal{O}\big((N-a-b)^2 d^2\big)$; fast convolution in t and fast polynomial multiplication in L reduce both factors. These are complexity $\mathcal{O}$'s and carry no analytic meaning. In practice, at the moderate orders typical of hand asymptotics, exact coefficient normalization — expansion, collection, and gcd reduction if the coefficients are rational functions of L — costs more than the convolution itself, and the decisive implementation choice is to never form a product whose outer order exceeds the requested cutoff.

Remark 1.17 (A rectangular cutoff is not a truncation). Discarding all monomials $t^n L^j$ with $|n|+|j|$ or $|j|$ above a threshold does not respect dominance. With a cap $|j|\leq 10$ one deletes t^2L^{100} while retaining t^3L^{-100}, although $t^3L^{-100}\prec t^2L^{100}$ by an enormous margin. Bounding $\deg_{\mathrm{L}} p_n$ is a second, independent approximation; it is legitimate only when accompanied by a proved degree bound such as Theorem 1.29.

1.3 Leading data and the leading-term law

Definition 1.18 (Leading data)**.** Let R be a logarithmically graded domain over $\mathbb{K}$ and let $F=\sum_{n\geq n_0} p_n t^n\in R((t))$ be nonzero with $\nu_t F=\min\{n : p_n\neq 0\}$; set $\nu_t 0=+\infty$. The *leading block* of F is the whole coefficient $p_{\nu_t F}\in R$, the *leading monomial* is

$$\mathrm{lm}(F):=t^{\nu_t F}L^{\deg_{\mathrm{L}} p_{\nu_t F}},\tag{15}$$

and the *leading scalar* is $\mathrm{lc}(F):=\mathrm{lc}_L\big(p_{\nu_t F}\big)\in\mathbb{K}^\times$. The *rank-two valuation* of F is

$$v(F):=\big(\nu_t F,\ -\deg_{\mathrm{L}} p_{\nu_t F}\big)\in\mathbb{Z}^2,\tag{16}$$

with $\mathbb{Z}^2$ lexicographically ordered and $v(0)=+\infty$. These are three different objects and none is determined by the coarse integer $\nu_t F$ alone.

Remark 1.19 (These are the objects of the ring chapter, over a general coefficient ring). Theorem 1.18 introduces no new object. It repeats **??** and **??** with the three coefficient rings $\mathbb{K}[L]$, $\mathbb{K}(L)$, $\mathbb{K}((L^{-1}))$ replaced by an arbitrary logarithmically graded domain R, so that the leading-term law can be proved once for all four ambient classes simultaneously; on those four classes the two readings agree. The same applies to Theorem 1.24 against **??** — whose codomain is $\mathbb{N}_0 \cup \{-\infty\}$ because there the blocks are polynomials, whereas over $\mathbb{K}(L)$ and $\mathbb{K}((L^{-1}))$ the value $\deg_{\mathrm{L}} p_n$ may be negative — and to Theorem 1.13 against **??**, which the corollary strengthens to arbitrary sufficient cutoffs (P, Q) and supplements with optimality and sharpness. Nothing below re-proves a ring-chapter result for its own sake; the restatements exist only to make the general coefficient ring available.

Lemma 1.20 (Dominance is growth)**.** *On the positive ray, for integers n, m and j, k,*

$$t^n L^j \succ t^m L^k \quad\iff\quad \lim_{X\to+\infty} \frac{X^{-n}(\log X)^j}{X^{-m}(\log X)^k} = +\infty \quad\iff\quad n < m,\ \text{or}\ (n = m\ \text{and}\ j > k).$$

Consequently, for F in $\mathscr{PL}_{\mathrm{poly},\mathbb{K}}$ or in $\mathscr{PL}_{\mathrm{itLaur},\mathbb{K}}$, $\mathrm{lm}(F)$ is the $\succ$-largest element of $\mathrm{supp}_{\mathrm{coeff}}(F)$. (For $\mathscr{PL}_{\mathrm{rat},\mathbb{K}}$ the statement is meaningful only after the expansion of Theorem 1.9.)

Proof. By **??** the relation $\succ$ is *defined* by the right-hand condition, so the first equivalence is a definition and only the second requires an argument; it is **??** specialized to integer exponents, and we repeat the two lines. The ratio equals $X^{m-n}(\log X)^{j-k}$. If $m > n$ it tends to $+\infty$ because $\log X = o_{X\to+\infty}(X^{\varepsilon})$ for every $\varepsilon > 0$; if $m = n$ it is $(\log X)^{j-k}$, which tends to $+\infty$ exactly when $j > k$; if $m < n$ it tends to 0 for the same reason as the first case. For the last sentence: a monomial $t^n L^j$ in the support has $n \geq \nu_t F$, with equality attained; among those with $n = \nu_t F$ the largest j with $\kappa_j(p_{\nu_t F}) \neq 0$ is by definition $\deg_{\mathrm{L}} p_{\nu_t F}$. $\square$

Theorem 1.21 (Leading-term law)**.** *Let R be a logarithmically graded domain over $\mathbb{K}$ and let $F, G \in R((t))$ be nonzero. Then $FG \neq 0$ and*

$$\nu_t(FG) = \nu_t F + \nu_t G, \tag{17}$$

$$\mathrm{lm}(FG) = \mathrm{lm}(F)\,\mathrm{lm}(G), \tag{18}$$

$$\mathrm{lc}(FG) = \mathrm{lc}(F)\,\mathrm{lc}(G), \tag{19}$$

$$v(FG) = v(F) + v(G). \tag{20}$$

Moreover $\nu_t(F + G) \geq \min(\nu_t F, \nu_t G)$ and $v(F + G) \geq \min\big(v(F), v(G)\big)$ in the lexicographic order. Each is an equality unless cancellation occurs at the common level: whole leading blocks cancel in the first case, top logarithmic coefficients in the second.

Proof. Put $a = \nu_t F$, $b = \nu_t G$. By Theorem 1.6 the coefficient C_N of FG vanishes for $N < a + b$, and for $N = a + b$ the index set $[a, N - b] = \{a\}$ is a single point, so $C_{a+b} = p_a q_b$. Since R is a domain (Theorem 1.3) and $p_a, q_b \neq 0$, we get $C_{a+b} \neq 0$; hence $FG \neq 0$ and (17) holds. Applying (2) to $C_{a+b} = p_a q_b$ gives

$$\deg_{\mathrm{L}} C_{a+b} = \deg_{\mathrm{L}} p_a + \deg_{\mathrm{L}} q_b, \qquad \mathrm{lc}_L(C_{a+b}) = \mathrm{lc}_L(p_a)\,\mathrm{lc}_L(q_b),$$

which are exactly (18) and (19); (20) is the same pair of identities rewritten in the notation (16).

For the sum, let $c = \min(a, b)$. Every block of $F + G$ of index $< c$ vanishes, so $\nu_t(F + G) \geq c$. If $a < b$ then $[t^c](F + G) = p_a \neq 0$ and the leading data of $F + G$ and F coincide, so $v(F + G) = v(F) = \min(v(F), v(G))$ — note that $a < b$ makes $v(F) < v(G)$ lexicographically. If $a = b = c$ and $p_c + q_c \neq 0$, then $\nu_t(F + G) = c$ and $\deg_{\mathrm{L}}(p_c + q_c) \leq \max(\deg_{\mathrm{L}} p_c, \deg_{\mathrm{L}} q_c)$ by (2), i.e. $-\deg_{\mathrm{L}}(p_c + q_c) \geq \min(-\deg_{\mathrm{L}} p_c, -\deg_{\mathrm{L}} q_c)$, which is $v(F + G) \geq \min(v(F), v(G))$. If $p_c + q_c = 0$, then $\nu_t(F + G) > c$ and $v(F + G) > \min(v(F), v(G))$ outright. $\square$

Corollary 1.22 (Integrality and fields). *Each of $\mathscr{PL}_{\mathrm{pow},\mathbb{K}}$, $\mathscr{PL}_{\mathrm{poly},\mathbb{K}}$, $\mathscr{PL}_{\mathrm{rat},\mathbb{K}}$, $\mathscr{PL}_{\mathrm{itLaur},\mathbb{K}}$ is an integral domain. The map v of (16) is a rank-two valuation on $\mathscr{PL}_{\mathrm{rat},\mathbb{K}}$ and on $\mathscr{PL}_{\mathrm{itLaur},\mathbb{K}}$, with value group $\mathbb{Z}^2$ lexicographically ordered; on $\mathscr{PL}_{\mathrm{poly},\mathbb{K}}$ it takes values in the sub-semigroup $\mathbb{Z}\times\mathbb{Z}_{\le 0}$. The classes $\mathscr{PL}_{\mathrm{rat},\mathbb{K}}$ and $\mathscr{PL}_{\mathrm{itLaur},\mathbb{K}}$ are fields; $\mathscr{PL}_{\mathrm{pow},\mathbb{K}}$ and $\mathscr{PL}_{\mathrm{poly},\mathbb{K}}$ are not.*

Proof. Integrality is Theorem 1.21, and the valuation axioms are its last two displays together with $v(F)=+\infty$ iff $F=0$. Surjectivity onto $\mathbb{Z}^2$ for $\mathscr{PL}_{\mathrm{rat},\mathbb{K}}$ and $\mathscr{PL}_{\mathrm{itLaur},\mathbb{K}}$ follows from $v(t^nL^{-j})=(n,j)$, legitimate there because L^{-j} lies in $\mathbb{K}(L)$ and in $\mathbb{K}((L^{-1}))$; the group $\mathbb{Z}^2$ with the lexicographic order has exactly one proper nontrivial convex subgroup, $\{0\}\times\mathbb{Z}$, so the valuation has rank two. On $\mathscr{PL}_{\mathrm{poly},\mathbb{K}}$ the second coordinate is $-\deg_{\mathrm{L}} p_{\nu_t F}\le 0$ because polynomial degrees are nonnegative.

A Laurent-series ring $R((t))$ over a *field* R is a field, which covers $\mathscr{PL}_{\mathrm{rat},\mathbb{K}}$ and $\mathscr{PL}_{\mathrm{itLaur},\mathbb{K}}$: given $F\neq 0$ with $\nu=\nu_t F$ and leading block $p_\nu\in R^\times$, write $F=p_\nu t^\nu(1+U)$ with $U=\sum_{n>\nu}(p_n/p_\nu)t^{n-\nu}\in tR[[t]]$; the family $\big((-U)^r\big)_{r\ge 0}$ is summable because $\nu_t(-U)^r\ge r$, and by Theorem 1.34(2) its sum V satisfies

$$(1+U)V=\sum_{r\ge 0}(-U)^r-\sum_{r\ge 0}(-U)^{r+1}=(-U)^0=1,$$

so $F^{-1}=p_\nu^{-1}t^{-\nu}V$. (Theorem 1.34 is proved in Section 1.5 from Theorem 1.21 alone, so there is no circularity.) Finally $L\in\mathscr{PL}_{\mathrm{poly},\mathbb{K}}$ is not invertible in $\mathscr{PL}_{\mathrm{poly},\mathbb{K}}$: if $LH=1$ then (18) gives $L\cdot\mathrm{lm}(H)=1$, forcing $\deg_{\mathrm{L}}$ of the leading block of H to be -1, impossible for a polynomial. Since $L\in\mathscr{PL}_{\mathrm{pow},\mathbb{K}}\subseteq\mathscr{PL}_{\mathrm{poly},\mathbb{K}}$, an inverse of L inside $\mathscr{PL}_{\mathrm{pow},\mathbb{K}}$ would in particular be an inverse inside $\mathscr{PL}_{\mathrm{poly},\mathbb{K}}$; so L is not invertible in $\mathscr{PL}_{\mathrm{pow},\mathbb{K}}$ either, and neither class is a field. □

Corollary 1.23 (Necessity half of the unit criterion). *If $F\in\mathscr{PL}_{\mathrm{poly},\mathbb{K}}$ is invertible in $\mathscr{PL}_{\mathrm{poly},\mathbb{K}}$, then its leading block is a nonzero scalar: $p_{\nu_t F}\in\mathbb{K}^\times$. The same statement holds in $\mathscr{PL}_{\mathrm{pow},\mathbb{K}}$ with the additional conclusion $\nu_t F=0$.*

Proof. If $FH=1$ then (17) gives $\nu_t F+\nu_t H=0$ and (18) gives $\deg_{\mathrm{L}} p_{\nu_t F}+\deg_{\mathrm{L}} h_{\nu_t H}=0$. Both degrees are nonnegative integers, hence both vanish, so $p_{\nu_t F}\in\mathbb{K}^\times$. In $\mathscr{PL}_{\mathrm{pow},\mathbb{K}}$ both orders are ≥ 0 and sum to 0, so both are 0. □

The converse — that a scalar leading block suffices — is a division statement and is proved with the reciprocal recurrence; it is not a multiplication fact and is not repeated here.

1.4 Degree and support bounds

Definition 1.24 (Log-degree profile). For $F=\sum_n p_n t^n$ in one of the four classes, the *log-degree profile* is the function $d_F\colon\mathbb{Z}\to\mathbb{Z}\cup\{-\infty\}$,

$$d_F(n):=\begin{cases}\deg_{\mathrm{L}} p_n, & p_n\neq 0,\\ -\infty, & p_n=0.\end{cases}\tag{21}$$

The profile is data additional to $\nu_t F$; it is not recoverable from it. For two profiles define the *max-plus convolution*

$$\big(d_F * d_G\big)(N):=\max_{n+m=N}\big(d_F(n)+d_G(m)\big),\tag{22}$$

with the conventions $-\infty+c=-\infty$ and $\max\emptyset=-\infty$. Only the finitely many pairs of Theorem 1.6 can contribute a value other than $-\infty$, so the maximum is effectively over a finite set.

Theorem 1.25 (Degree bound and exact criterion for attainment). *Let $F,G\in R((t))$ with R a logarithmically graded domain over $\mathbb{K}$, let $N\in\mathbb{Z}$, and set $M:=(d_F*d_G)(N)$. Let*

$$P_N:=\{(n,m):n+m=N,\ p_n\neq 0,\ q_m\neq 0,\ d_F(n)+d_G(m)=M\}$$

be the (finite, possibly empty) set of maximizing pairs. Then

$$d_{FG}(N) \le M, \tag{23}$$

If $P_N = \emptyset$, equivalently $M = -\infty$, then $[t^N](FG) = 0$ and (23) holds with equality. If $P_N \ne \emptyset$, so that $M \in \mathbb{Z}$, then the coefficient of L^M in the block $[t^N](FG)$ is exactly

$$\kappa_M\big([t^N](FG)\big) = \sum_{(n,m)\in P_N} \mathrm{lc}_L(p_n)\, \mathrm{lc}_L(q_m), \tag{24}$$

and equality holds in (23) if and only if the sum (24) is nonzero.

Proof. Each summand of $C_N = \sum_{n+m=N} p_n q_m$ has $\deg_{\mathrm{L}}(p_n q_m) = d_F(n) + d_G(m) \le M$ by (2), with the convention that a vanishing factor gives $-\infty$. A finite sum of elements of $\deg_{\mathrm{L}} \le M$ has $\deg_{\mathrm{L}} \le M$, which is (23). If $M = -\infty$ then every summand $p_n q_m$ vanishes, so $[t^N](FG) = 0$ and both sides of (23) equal $-\infty$; assume from now on $P_N \ne \emptyset$, so that $M \in \mathbb{Z}$. By $\mathbb{K}$-linearity of κ_M,

$$\kappa_M(C_N) = \sum_{n+m=N} \kappa_M(p_n q_m).$$

If $(n, m) \notin P_N$ then either a factor vanishes or $\deg_{\mathrm{L}}(p_n q_m) < M$, and in both cases $\kappa_M(p_n q_m) = 0$. If $(n, m) \in P_N$ then $\deg_{\mathrm{L}}(p_n q_m) = M$ and $\kappa_M(p_n q_m) = \mathrm{lc}_L(p_n q_m) = \mathrm{lc}_L(p_n)\,\mathrm{lc}_L(q_m)$ by Theorem 1.3. This proves (24), and equality in (23) means precisely $\kappa_M(C_N) \ne 0$. □

Corollary 1.26 (Three sufficient conditions for equality)**.** *In the situation of Theorem 1.25, equality holds in (23) in each of the following cases.*

1. *P_N is a singleton.*
2. *$N = \nu_t F + \nu_t G$ (both factors nonzero); then P_N is the singleton $\{(\nu_t F, \nu_t G)\}$ and one recovers (18).*
3. *$P_N \ne \emptyset$, $\mathbb{K}$ is an ordered field, and $\mathrm{lc}_L(p_n) > 0$, $\mathrm{lc}_L(q_m) > 0$ for every $(n, m) \in P_N$.*

Proof. (1) The sum (24) is a single product of two elements of $\mathbb{K}^\times$, hence nonzero. (2) By Theorem 1.6 the only admissible pair at $N = \nu_t F + \nu_t G$ is $(\nu_t F, \nu_t G)$, and both blocks are nonzero, so P_N is that singleton; apply (1). (3) The sum is a nonempty sum of strictly positive elements of an ordered field. □

Example 1.27 (The bound is attained, at every order and by every pair)**.** Over $\mathbb{K} = \mathbb{Q}$ let

$$F = G = \sum_{n\ge 0} L^n t^n \in \mathscr{PL}_{\mathrm{pow},\mathbb{K}}, \qquad \text{so} \qquad d_F(n) = d_G(n) = n.$$

Then $(d_F * d_G)(N) = \max_{n+m=N}(n + m) = N$, and *every* one of the $N + 1$ admissible pairs is maximizing, so P_N is as large as it can be and Theorem 1.26(1) does not apply. Nevertheless

$$[t^N](FG) = \sum_{n=0}^{N} L^n \cdot L^{N-n} = (N + 1)L^N,$$

so $d_{FG}(N) = N$ for every N: the bound (23) is attained at every order. Formula (24) predicts exactly the top coefficient $N + 1 = \sum_{(n,m)\in P_N} 1 \cdot 1$.

A common trap

Theorem 1.27 is the place where characteristic zero is used. Over a field of characteristic $p > 0$ the same computation gives $[t^{p-1}](FG) = p\,L^{p-1} = 0$, so the block vanishes entirely and $d_{FG}(p - 1) = -\infty$ although the bound is $p - 1$. Every statement in this section is asserted over a field of characteristic zero, and this is not a formality.

Example 1.28 (The bound can fail by an arbitrary amount)**.** Let $F = 1 + L^d t$ and $G = 1 - L^d t$ in $\mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}}$, $d \geq 1$. Then $(d_F * d_G)(1) = \max(0 + d,\ d + 0) = d$, while $[t^1](FG) = -L^d + L^d = 0$, so $d_{FG}(1) = -\infty$. Here $P_1 = \{(0,1),(1,0)\}$ and the sum (24) is $1 \cdot (-1) + 1 \cdot 1 = 0$, as it must be. A drop by exactly one occurs for $F = 1 + (L^2 + L)t$, $G = 1 - (L^2 - L)t$, where the bound at $N = 1$ is 2 while $[t^1](FG) = 2L$ has degree 1: here the top coefficients cancel but the next ones do not.

Proposition 1.29 (Affine profiles and the profile subalgebra)**.** *Let $\alpha, \alpha' \geq 0$ and $\beta, \gamma \in \mathbb{R}$. Let $F, G \in \mathscr{P}\mathscr{L}_{\mathrm{poly},\mathbb{K}}$ be nonzero with $a = \nu_t F$, $b = \nu_t G$ and*

$$d_F(n) \leq \alpha n + \beta \ \ (n \geq a), \qquad d_G(m) \leq \alpha' m + \gamma \ \ (m \geq b).$$

Put $\mu = \max(\alpha, \alpha')$. Then for all N,

$$d_{FG}(N) \leq \mu N + \beta + \gamma - (\mu - \alpha)a - (\mu - \alpha')b. \tag{25}$$

In particular, if $a, b \geq 0$ then $d_{FG}(N) \leq \mu N + \beta + \gamma$. Consequently, for each $\alpha \geq 0$ the set

$$\mathfrak{A}_\alpha := \{F \in \mathscr{P}\mathscr{L}_{\mathrm{poly},\mathbb{K}} : \ \exists \beta \in \mathbb{R},\ d_F(n) \leq \alpha n + \beta \text{ for all } n\}$$

is a $\mathbb{K}$-subspace with $\mathfrak{A}_\alpha \cdot \mathfrak{A}_{\alpha'} \subseteq \mathfrak{A}_{\max(\alpha,\alpha')}$, and $\mathfrak{A} := \bigcup_{\alpha \geq 0} \mathfrak{A}_\alpha$ is a $\mathbb{K}$-subalgebra of $\mathscr{P}\mathscr{L}_{\mathrm{poly},\mathbb{K}}$ containing $\mathbb{K}[L][t, t^{-1}]$.

Proof. Fix N and $n + m = N$ with $n \geq a$, $m \geq b$. Then

$$\alpha n + \alpha' m = \mu(n + m) - (\mu - \alpha)n - (\mu - \alpha')m \leq \mu N - (\mu - \alpha)a - (\mu - \alpha')b,$$

because $\mu - \alpha \geq 0$, $\mu - \alpha' \geq 0$, $n \geq a$ and $m \geq b$. Adding $\beta + \gamma$ and taking the maximum over the finite index set gives (25) via (23). If $a, b \geq 0$ then the two subtracted terms are ≥ 0, so they may be dropped.

For the algebra statements: $\mathfrak{A}_\alpha$ is a subspace because $d_{F+G}(n) \leq \max(d_F(n), d_G(n)) \leq \alpha n + \max(\beta, \gamma)$ and scalars do not change profiles. The product statement is (25), whose right-hand side is $\mu N + \text{const}$. For $\alpha \leq \alpha''$ one has $\mathfrak{A}_\alpha \subseteq \mathfrak{A}_{\alpha''}$. Indeed, let $F \neq 0$ satisfy $d_F(n) \leq \alpha n + \beta$ for all n, and put $a = \nu_t F$. For $n \geq 0$ we have $\alpha n + \beta \leq \alpha'' n + \beta$; for $a \leq n < 0$,

$$\alpha n + \beta = \alpha'' n + (\alpha'' - \alpha)(-n) + \beta \leq \alpha'' n + \beta + (\alpha'' - \alpha)\max(0, -a);$$

and $d_F(n) = -\infty$ for $n < a$. Hence $\beta'' = \beta + (\alpha'' - \alpha)\max(0, -a)$ witnesses $F \in \mathfrak{A}_{\alpha''}$, and $0 \in \mathfrak{A}_{\alpha''}$ trivially. Consequently the union $\mathfrak{A}$ is closed under addition and multiplication and contains 1; every Laurent polynomial $\sum_{n=n_1}^{n_2} p_n(L)t^n$ lies in $\mathfrak{A}_0$ with $\beta = \max_n \deg_{\mathrm{L}} p_n$. ☐

Remark 1.30 ($\mathfrak{A}$ is not t-adically closed). The affine-profile condition is preserved by finite sums and products but not by t-adic limits. Take $F_r = t^r L^{r^2}$ for $r \geq 1$; each F_r lies in $\mathfrak{A}_0$, and $\nu_t F_r = r \to \infty$, so (F_r) is summable, yet the sum $S = \sum_{r \geq 1} t^r L^{r^2}$ has $d_S(r) = r^2$, which is bounded by no affine function. Hence $S \notin \mathfrak{A}$. A degree bound must therefore be re-derived, not inherited, whenever an infinite construction is performed.

Proposition 1.31 (Profile of a power)**.** *Let $F \in \mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}}$ with $d_F(n) \leq \alpha n + \beta$ for all $n \geq 0$, where $\alpha \geq 0$ and $\beta \geq 0$. Then for every integer $k \geq 0$,*

$$d_{F^k}(N) \leq \alpha N + k\beta \qquad (N \geq 0). \tag{26}$$

Proof. Induction on k. For $k = 0$ we have $F^0 = 1$, so $d_{F^0}(0) = 0 \leq 0$ and $d_{F^0}(N) = -\infty$ for $N > 0$. Assume the bound for k. By (23) applied to $F^{k+1} = F^k \cdot F$,

$$d_{F^{k+1}}(N) \leq \max_{n+m=N,\ n,m \geq 0} \big((\alpha n + k\beta) + (\alpha m + \beta)\big) = \alpha N + (k+1)\beta.$$

☐

Remark 1.32 (The Lambert profile, corrected). Two of the sources record the degree law of the canonical Lambert family as $\deg \mathsf{L}_n^{\mathrm{Lam}} = n$ without restriction, and use it to assert that products of Lambert-type expansions have profile $\leq N$. The law $\deg \mathsf{L}_n^{\mathrm{Lam}} = n$ holds only for $n \geq 1$: the normalization $\mathsf{L}_0^{\mathrm{Lam}}(u) = -u$ has degree 1, not 0. The correct hypothesis for a series $F = \sum_{n\geq 0} \mathsf{L}_n^{\mathrm{Lam}}(L)t^n$ is therefore $d_F(n) \leq n+1$ for all $n \geq 0$, i.e. $\alpha = 1$, $\beta = 1$ in Theorem 1.29; a product of two such series has $d_{FG}(N) \leq N+2$, and a k-th power has $d_{F^k}(N) \leq N+k$ by Theorem 1.31. The unrestricted claim $d_{FG}(N) \leq N$ is false already at $N = 0$, where $[t^0](FG) = \mathsf{L}_0^{\mathrm{Lam}}(L)^2 = L^2$ has degree 2.

1.5 Powers and binomial powers

Infinite sums enter through powers, so we first record the exact sense in which they are legitimate.

Definition 1.33 (Summable family)**.** Let R be a logarithmically graded domain over $\mathbb{K}$. A family $(S_i)_{i\in I}$ in $R((t))$ is *summable* if for every $N \in \mathbb{Z}$ the set $\{i \in I : \nu_t S_i < N\}$ is finite. Its sum is defined blockwise: $[t^n]\sum_i S_i := \sum_i [t^n]S_i$, a finite sum in the coefficient ring.

Lemma 1.34 (Summable families are well behaved)**.** *Let $(S_i)_{i\in I}$ and $(T_j)_{j\in J}$ be summable families in $R((t))$.*

1. $\inf_i \nu_t S_i > -\infty$*, so* $\sum_i S_i$ *is a genuine element of* $R((t))$ *(its support is bounded below).*
2. *The family* $(S_iT_j)_{(i,j)\in I\times J}$ *is summable and* $\big(\sum_i S_i\big)\big(\sum_j T_j\big) = \sum_{(i,j)} S_iT_j$.

Proof. (1) Taking $N = 0$ in Theorem 1.33, only finitely many i have $\nu_t S_i < 0$, and each of those has $\nu_t S_i \in \mathbb{Z}$ (a member of that finite set is in particular nonzero). Let λ be the minimum of that finite set of integers, or $\lambda = 0$ if the set is empty. Then $\nu_t S_i \geq \min(0, \lambda)$ for all i, and every block of $\sum_i S_i$ below that bound vanishes.

(2) Let $\alpha = \inf_i \nu_t S_i$ and $\beta = \inf_j \nu_t T_j$, both $> -\infty$ by (1). If $\alpha = +\infty$ or $\beta = +\infty$ then every S_i, respectively every T_j, is zero, so every S_iT_j is zero and both assertions are trivial; assume therefore $\alpha, \beta \in \mathbb{Z}$. Fix N. If $\nu_t(S_iT_j) = \nu_t S_i + \nu_t T_j < N$ then $\nu_t S_i < N - \beta$ and $\nu_t T_j < N - \alpha$, and each of those conditions holds for only finitely many indices; so the family is summable. For the identity, fix n and compute in the coefficient ring, all sums having only finitely many nonzero terms:

$$[t^n]\Big(\sum_i S_i \sum_j T_j\Big) = \sum_{p+q=n} \Big(\sum_i [t^p]S_i\Big)\Big(\sum_j [t^q]T_j\Big) = \sum_{i,j}\sum_{p+q=n} [t^p]S_i\,[t^q]T_j = \sum_{i,j}[t^n](S_iT_j).$$

□

Remark 1.35. Part (1) is worth isolating. For a *sequence* in a Laurent ring, t-adic Cauchyness alone does not guarantee a limit with a finite principal part; a common eventual lower support bound must be imposed. For a *summable family* no such extra hypothesis is needed, because the finiteness condition at $N = 0$ already forces a common lower bound.

Proposition 1.36 (Coefficients of an integer power)**.** *Let $F = \sum_{n\geq 0} p_n t^n \in \mathscr{PL}_{\mathrm{pow},\mathbb{K}}$ and $k \geq 0$. Then*

$$[t^N]F^k = \sum_{\substack{n_1,\ldots,n_k\geq 0\\ n_1+\cdots+n_k=N}} p_{n_1}\cdots p_{n_k} := C_{N,k}(p_0, p_1, \ldots), \tag{27}$$

with $C_{0,0} = 1$ and $C_{N,0} = 0$ for $N > 0$. These convolution polynomials *satisfy*

$$C_{N,k+1} = \sum_{n=0}^{N} p_n\, C_{N-n,k}, \tag{28}$$

which computes the whole triangular table in $\mathcal{O}(N^2k)$ coefficient products. If $p_0 = 0$, then $C_{N,k} = 0$ for $N < k$, and, writing $F = \sum_{n\geq 1} p_n t^n$,

$$[t^N]F^k = \frac{k!}{N!}\,\mathcal{B}^{\exp}_{N,k}\big(1!\,p_1,\ 2!\,p_2, \ldots, (N-k+1)!\,p_{N-k+1}\big). \tag{29}$$

Proof. Formula (27) follows by induction on k from Theorem 1.5: the case $k=0$ is the convention, and

$$[t^N]F^{k+1} = \sum_{n+m=N} p_n\,[t^m]F^k = \sum_{n=0}^{N} p_n \sum_{\substack{n_1,\dots,n_k\ge 0\\ n_1+\dots+n_k=N-n}} p_{n_1}\cdots p_{n_k},$$

which is both (28) and the composition sum for $k+1$. All sums are finite because every $n_i \ge 0$ and they sum to N. If $p_0 = 0$, every n_i in a nonvanishing term is ≥ 1, so $N \ge k$.

For (29), recall the defining generating identity of the exponential partial Bell polynomials [?, ?]: in $R[[z]]$ over a commutative $\mathbb{Q}$-algebra R,

$$\frac{1}{k!}\Big(\sum_{j\ge 1} x_j \frac{z^j}{j!}\Big)^k = \sum_{N\ge k} \mathcal{B}^{\exp}_{N,k}(x_1, x_2, \dots)\frac{z^N}{N!}.$$

Apply it with $R = \mathbb{K}[L]$ (a $\mathbb{Q}$-algebra, as $\operatorname{char}\mathbb{K} = 0$), $z = t$ and $x_j = j!\,p_j$, so that $\sum_{j\ge 1} x_j z^j/j! = F$. Comparing coefficients of t^N gives $\frac{1}{k!}[t^N]F^k = \frac{1}{N!}\mathcal{B}^{\exp}_{N,k}(1!p_1, 2!p_2, \dots)$, which is (29); only the arguments $x_1, \dots, x_{N-k+1}$ occur because a term of $\mathcal{B}^{\exp}_{N,k}$ is a product of k variables with index sum N, so no index can exceed $N-k+1$. □

Theorem 1.37 (Binomial powers and their finite determination). *Let $U \in \mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}}$ with $\nu_t U \ge e$ for an integer $e \ge 1$, write $u_n = [t^n]U$, and let $\sigma \in \mathbb{K}$. Put $\binom{\sigma}{r} = \sigma(\sigma-1)\cdots(\sigma-r+1)/r! \in \mathbb{K}$, with $\binom{\sigma}{0} = 1$. Then:*

1. *The family $\big(\binom{\sigma}{r}U^r\big)_{r\ge 0}$ is summable, so*
$$(1+U)^\sigma := \sum_{r\ge 0}\binom{\sigma}{r}U^r \tag{30}$$
is a well-defined element of $\mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}}$ with constant block 1.

2. (Finite determination.) *For $N \ge 0$,*
$$[t^N](1+U)^\sigma = \sum_{r=0}^{\lfloor N/e\rfloor}\binom{\sigma}{r}\,[t^N]U^r, \tag{31}$$
and this depends only on the blocks $u_e, u_{e+1}, \dots, u_N$, that is, only on $\operatorname{Trunc}_{<N+1} U$. The range of r is exact: for $U = t^e$ with $e \mid N$ the term $r = N/e$ contributes $\binom{\sigma}{N/e}$, nonzero whenever $\sigma \notin \{0, 1, \dots, N/e - 1\}$. The top block enters exactly linearly: for $N \ge 1$,
$$[t^N](1+U)^\sigma = \sigma u_N + \Psi_N(u_e, \dots, u_{N-e}), \tag{32}$$
where Ψ_N is a universal polynomial with coefficients in $\mathbb{Q}[\sigma]$, with the convention that $\Psi_N = 0$ when the argument list is empty. Hence for $\sigma \ne 0$ the block u_N cannot be omitted from the dependency range.

3. (Exponent law.) *For $\sigma, \tau \in \mathbb{K}$,*
$$(1+U)^\sigma(1+U)^\tau = (1+U)^{\sigma+\tau}, \qquad (1+U)^0 = 1, \tag{33}$$
and for $k \in \mathbb{N}_0$ the series $(1+U)^k$ agrees with the k-fold product. Consequently $(1+U)^\sigma$ is a unit of $\mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}}$ with inverse $(1+U)^{-\sigma}$, and $\sigma \mapsto (1+U)^\sigma$ is a homomorphism from $(\mathbb{K}, +)$ to the units of $\mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}}$.

4. (Profile.) *If $d_U(n) \le \alpha n + \beta$ for all $n \ge e$, with $\alpha \ge 0$, then, writing $\beta^+ = \max(\beta, 0)$,*
$$d_{(1+U)^\sigma}(N) \le \Big(\alpha + \frac{\beta^+}{e}\Big)N \qquad (N \ge 0). \tag{34}$$
In particular $\mathfrak{A} \cap \mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}}$ is stable under $U \mapsto (1+U)^\sigma$ whenever $\nu_t U \ge 1$.

Proof. (1) By (17), $\nu_t(U^r) \geq re$, which tends to $+\infty$; so for each N only the finitely many $r \leq N/e$ can have $\nu_t(\binom{\sigma}{r}U^r) < N$. Theorem 1.34(1) gives a well-defined sum, and $\nu_t U \geq e \geq 1$ makes the $r = 0$ term the only contributor to t^0.

(2) Blockwise, $[t^N](1+U)^\sigma = \sum_{r\geq 0}\binom{\sigma}{r}[t^N]U^r$, and $[t^N]U^r = 0$ once $re > N$; this is (31). By Theorem 1.12 applied $r-1$ times, $[t^N]U^r$ depends only on the blocks u_n with $e \leq n \leq N-(r-1)e$, and for $r \geq 1$ this range is contained in $[e, N]$. For the sharpness of the r-range take $U = t^e$: then $U^r = t^{re}$ and the only contribution to t^N comes from $r = N/e$, with value $\binom{\sigma}{N/e}$, which vanishes exactly when σ is one of $0, 1, \ldots, N/e - 1$. For (32), the term $r = 0$ contributes nothing to t^N for $N \geq 1$, the term $r = 1$ contributes σu_N, and each term with $r \geq 2$ involves only u_n with $n \leq N - (r-1)e \leq N - e$.

(3) By Theorem 1.34(2),

$$(1+U)^\sigma(1+U)^\tau = \sum_{r,s\geq 0}\binom{\sigma}{r}\binom{\tau}{s}U^{r+s} = \sum_{m\geq 0}\Big(\sum_{r+s=m}\binom{\sigma}{r}\binom{\tau}{s}\Big)U^m,$$

the regrouping being legitimate because for each fixed t-block only finitely many pairs (r, s) contribute. Vandermonde's convolution [?]

$$\sum_{r+s=m}\binom{\sigma}{r}\binom{\tau}{s} = \binom{\sigma+\tau}{m}$$

holds for all $\sigma, \tau \in \mathbb{K}$: both sides are polynomial expressions in (σ, τ) with coefficients in $\mathbb{Q}$; they agree for all pairs of nonnegative integers by comparing coefficients in $(1+z)^\sigma(1+z)^\tau = (1+z)^{\sigma+\tau}$ in $\mathbb{Q}[z]$; a polynomial in two variables over the infinite domain $\mathbb{Q}$ vanishing on $\mathbb{N}_0^2$ is the zero polynomial; and $\mathbb{Q} \subseteq \mathbb{K}$ because $\operatorname{char}\mathbb{K} = 0$, so the identity persists in $\mathbb{K}$. This proves (33); $(1+U)^0 = 1$ is immediate. For $k \in \mathbb{N}_0$ we have $\binom{k}{r} = 0$ for $r > k$, so (30) is the finite binomial expansion of the k-fold product in the commutative ring $\mathscr{PL}_{\mathrm{pow},\mathbb{K}}$. Invertibility follows by taking $\tau = -\sigma$.

(4) By (31) it suffices to bound $\deg_{\mathrm{L}}[t^N]U^r$ for $1 \leq r \leq \lfloor N/e\rfloor$ (the term $r = 0$ contributes only at $N = 0$, where the bound reads $0 \leq 0$). By (27) and (2),

$$\deg_{\mathrm{L}}[t^N]U^r \leq \max_{\substack{n_1+\cdots+n_r=N\\ n_i\geq e}}\sum_{i=1}^{r}(\alpha n_i + \beta) = \alpha N + r\beta \leq \alpha N + \lfloor N/e\rfloor\,\beta^+ \leq \Big(\alpha + \frac{\beta^+}{e}\Big)N,$$

where the second inequality uses $r \leq \lfloor N/e\rfloor$ when $\beta \geq 0$ and $r\beta \leq 0 \leq \lfloor N/e\rfloor\,\beta^+$ when $\beta < 0$. Taking the maximum over r gives (34). □

Corollary 1.38 (Formal roots)**.** *With U as above and $p, q \in \mathbb{Z}$, $q \geq 1$,*

$$\big((1+U)^{p/q}\big)^q = (1+U)^p,$$

the right-hand side being the ordinary p-th power (or the reciprocal of the $|p|$-th power if $p < 0$). In particular $(1+U)^{1/q}$ is a q-th root of $1+U$ in $\mathscr{PL}_{\mathrm{pow},\mathbb{K}}$, and it is the unique one with constant block 1.

Proof. Iterating (33) $q-1$ times gives $((1+U)^{p/q})^q = (1+U)^{qp/q} = (1+U)^p$, and Theorem 1.37(3) identifies the right-hand side with the ordinary power. For uniqueness, suppose $V, W \in \mathscr{PL}_{\mathrm{pow},\mathbb{K}}$ have constant block 1 and $V^q = W^q = 1+U$. Then $(V-W)\sum_{i=0}^{q-1}V^iW^{q-1-i} = 0$; the second factor has constant block $q \neq 0$ (characteristic zero), hence is nonzero, and $\mathscr{PL}_{\mathrm{pow},\mathbb{K}}$ is a domain by Theorem 1.22, so $V = W$. □

Remark 1.39 (Infinite products). If $(U_r)_{r\geq 1}$ lies in $\mathscr{PL}_{\mathrm{pow},\mathbb{K}}$ with $\nu_t U_r \to +\infty$, the partial products $P_R = \prod_{r\leq R}(1+U_r)$ converge t-adically. Indeed, given N choose R_N with $\nu_t U_r \geq N$ for $r > R_N$; for $R > R_N$,

$$P_R - P_{R_N} = P_{R_N}\Big(\prod_{R_N<r\leq R}(1+U_r) - 1\Big),$$

and expanding the finite product shows that the bracket is a sum of products of at least one U_r with $r > R_N$, hence of $\nu_t \geq N$; since $\nu_t P_{R_N} \geq 0$ we get $\nu_t(P_R - P_{R_N}) \geq N$. Thus (P_R) is t-adically Cauchy in the complete ring $\mathscr{PL}_{\mathrm{pow},\mathbb{K}}$ and $\prod_{r\geq 1}(1+U_r)$ is well defined. On $\mathscr{PL}_{\mathrm{poly},\mathbb{K}}$ the same argument applies once one notes that $\nu_t U_r \to +\infty$ forces all but finitely many factors to lie in $\mathscr{PL}_{\mathrm{pow},\mathbb{K}}$, so the partial products have a common lower support bound.

1.6 Three worked computations

Example 1.40 (A Laurent product through four blocks, with degree bookkeeping)**.** Over $\mathbb{K} = \mathbb{Q}$ take

$$F = 3t^{-2} + (L-1)t^{-1} + (L^2+2) + 2L\,t + (L^2-L)t^2 + \mathcal{O}_t^{\mathrm{form}}(t^3)\,,$$
$$G = t^{-1} + (2-L) + (L+1)t + L^2t^2 + \mathcal{O}_t^{\mathrm{form}}(t^3)\,.$$

Here $\nu_t F = -2$ and $\nu_t G = -1$; F is known through $\mathrm{Trunc}_{<3} F$ and G through $\mathrm{Trunc}_{<3} G$. By the precision rule the product is known through

$$\mathrm{Trunc}_{<\min(3+(-1),\,3+(-2))} FG = \mathrm{Trunc}_{<1} FG,$$

that is, through the block t^0 inclusive — four blocks in all, starting at t^{-3}. The extra block $[t^2]F$ is wasted: it can influence only $[t^1](FG)$ and beyond, for which $[t^3]G$ would also be needed.

Writing $A_n = [t^n]F$ and $B_m = [t^m]G$, the convolution (5) gives

$$\begin{aligned}
C_{-3} &= A_{-2}B_{-1} = 3,\\
C_{-2} &= A_{-2}B_0 + A_{-1}B_{-1} = 3(2-L) + (L-1) = 5-2L,\\
C_{-1} &= A_{-2}B_1 + A_{-1}B_0 + A_0B_{-1}\\
&= 3(L+1) + (-L^2+3L-2) + (L^2+2) = 6L+3,\\
C_0 &= A_{-2}B_2 + A_{-1}B_1 + A_0B_0 + A_1B_{-1}\\
&= 3L^2 + (L^2-1) + (-L^3+2L^2-2L+4) + 2L = -L^3+6L^2+3.
\end{aligned}$$

Hence

$$FG = 3t^{-3} + (5-2L)t^{-2} + (6L+3)t^{-1} + (-L^3+6L^2+3) + \mathcal{O}_t^{\mathrm{form}}(t^1)\,. \tag{35}$$

Leading data: $\mathrm{lm}(F) = t^{-2}$ with $\mathrm{lc}(F) = 3$, $\mathrm{lm}(G) = t^{-1}$ with $\mathrm{lc}(G) = 1$; Theorem 1.21 predicts $\mathrm{lm}(FG) = t^{-3}$ and $\mathrm{lc}(FG) = 3$, confirmed by $C_{-3} = 3$.

The degree bookkeeping is instructive. The profiles are $d_F(-2) = 0$, $d_F(-1) = 1$, $d_F(0) = 2$, $d_F(1) = 1$ and $d_G(-1) = 0$, $d_G(0) = 1$, $d_G(1) = 1$, $d_G(2) = 2$. Then:

N	maximizing pairs P_N	bound M	top coefficient (24)	$d_{FG}(N)$
-3	$(-2,-1)$	0	$3\cdot 1 = 3$	0
-2	$(-2,0),\ (-1,-1)$	1	$3(-1)+1\cdot 1 = -2$	1
-1	$(-1,0),\ (0,-1)$	2	$1\cdot(-1)+1\cdot 1 = 0$	$1<2$
0	$(0,0)$	3	$1\cdot(-1) = -1$	3

At $N=-2$ two pairs compete and their contributions -3 and $+1$ do not cancel, so the bound is attained, with top coefficient -2: indeed $C_{-2} = -2L+5$. At $N=-1$ two pairs compete and cancel exactly, so the degree drops from the bound 2 to the actual value 1: indeed $C_{-1} = 6L+3$. At $N=0$ the maximizing pair is unique, so Theorem 1.26(1) forces attainment, with top coefficient -1: indeed $C_0 = -L^3 + \cdots$. This is exactly the phenomenon Theorem 1.25 isolates; no inspection of the finished polynomial is needed to predict it.

Example 1.41 (A square root through four blocks, verified by squaring)**.** Let

$$U = 2Lt + (L^2+2)t^2 + 4L\,t^3 + \mathcal{O}_t^{\mathrm{form}}(t^4) \in t\mathscr{PL}_{\mathrm{pow},\mathbb{K}},$$

so $e = 1$ in Theorem 1.37, and take $\sigma = \frac{1}{2}$, with

$$\binom{1/2}{0} = 1, \quad \binom{1/2}{1} = \tfrac{1}{2}, \quad \binom{1/2}{2} = -\tfrac{1}{8}, \quad \binom{1/2}{3} = \tfrac{1}{16}.$$

By (31) only $r \le N$ contributes to t^N, so through t^3 we need

$$U^2 = 4L^2t^2 + (4L^3 + 8L)t^3 + \mathcal{O}_t^{\mathrm{form}}(t^4)\,, \qquad U^3 = 8L^3t^3 + \mathcal{O}_t^{\mathrm{form}}(t^4)\,.$$

Block by block:

$$\begin{aligned}
[t^1](1+U)^{1/2} &= \tfrac{1}{2}(2L) = L,\\
[t^2](1+U)^{1/2} &= \tfrac{1}{2}(L^2+2) - \tfrac{1}{8}(4L^2) = \tfrac{L^2}{2} + 1 - \tfrac{L^2}{2} = 1,\\
[t^3](1+U)^{1/2} &= \tfrac{1}{2}(4L) - \tfrac{1}{8}(4L^3+8L) + \tfrac{1}{16}(8L^3) = 2L - \tfrac{L^3}{2} - L + \tfrac{L^3}{2} = L.
\end{aligned}$$

Hence

$$(1+U)^{1/2} = 1 + L\,t + t^2 + L\,t^3 + \mathcal{O}_t^{\mathrm{form}}(t^4)\,. \tag{36}$$

Two logarithmic cancellations occur, one at each of t^2 and t^3; the degree bound (34) with $\alpha = 1$, $\beta^+ = 0$, $e = 1$ predicts only $d_{(1+U)^{1/2}}(N) \le N$, and the actual profile $0, 1, 0, 1$ is strictly smaller at $N = 2, 3$. A degree bound is an upper bound; it is not a prediction.

Verification. Theorem 1.38 says the square of (36) must reproduce $1 + U$ through t^3. Squaring by (5):

$$[t^0] = 1, \quad [t^1] = 2L, \quad [t^2] = 2 \cdot 1 + L^2 = L^2 + 2, \quad [t^3] = 2L + 2L = 4L,$$

which are exactly the blocks $1, u_1, u_2, u_3$ of $1 + U$. Note the precision accounting: squaring a series known through $\mathrm{Trunc}_{<4}\cdot$ with $\nu_t = 0$ returns a result known through $\mathrm{Trunc}_{<4}\cdot$, so the check is available at every block computed and at none beyond.

Example 1.42 (A product in the iterated field)**.** In $\mathscr{PL}_{\mathrm{itLaur},\mathbb{K}}$ the coefficient convolution is finite at the inner level too. Take the two blocks

$$f = \frac{1}{L+1} = L^{-1} - L^{-2} + L^{-3} - \cdots\,, \qquad g = \frac{1}{L-1} = L^{-1} + L^{-2} + L^{-3} + \cdots$$

in $\mathbb{K}((L^{-1}))$. For $r \ge 2$, the coefficient of L^{-r} in fg is a sum over the finitely many decompositions $r = i + j$ with $i, j \ge 1$:

$$\kappa_{-r}(fg) = \sum_{i=1}^{r-1} (-1)^{i+1} \cdot 1 = \begin{cases} 1, & r \text{ even},\\ 0, & r \text{ odd}, \end{cases}$$

so $fg = L^{-2} + L^{-4} + L^{-6} + \cdots = (L^2-1)^{-1}$, as it must be. The degree law is visible: $\deg_{\mathrm{L}} f = \deg_{\mathrm{L}} g = -1$, $\deg_{\mathrm{L}}(fg) = -2$, and $\mathrm{lc}_L(f) = \mathrm{lc}_L(g) = \mathrm{lc}_L(fg) = 1$. Consequently, for $F = f\,t^{-1} + \cdots$ and $G = g + \cdots$ in $\mathscr{PL}_{\mathrm{itLaur},\mathbb{K}}$ one gets $\mathrm{lm}(FG) = t^{-1}L^{-2}$ and $\mathrm{lc}(FG) = 1$ by Theorem 1.21, with no need to compute any further block. Observe that lm here carries a *negative* power of L: this is legitimate in $\mathscr{PL}_{\mathrm{rat},\mathbb{K}}$ and $\mathscr{PL}_{\mathrm{itLaur},\mathbb{K}}$ and impossible in $\mathscr{PL}_{\mathrm{poly},\mathbb{K}}$, which is exactly the content of Theorem 1.22.

1.7 The analytic product rule

Everything above is formal algebra. We now record the exact circumstances under which the block convolution computes the asymptotic expansion of an actual product of functions. Nothing in this subsection asserts convergence: an asymptotic expansion is a statement about remainders, and the formal product theorem is used as an input, not replaced.

Definition 1.43 (Polynomial–logarithmic asymptotic expansion). Let $\mathbb{K} \subseteq \mathbb{C}$, let $X_0 > 1$, and let $S = [X_0, +\infty)$ with the real branch $L = \log X$. A function $f\colon S \to \mathbb{C}$ is *PL-asymptotic* to $F \in \mathscr{PL}_{\mathrm{poly},\mathbb{K}}$, written $f \approx F$, if for every $N \in \mathbb{Z}$ there exist $M_N \in \mathbb{N}_0$, $C_N > 0$ and $X_N \geq X_0$ such that

$$\big|f(X) - \big(\mathrm{Trunc}_{<N} F\big)(X)\big| \leq C_N\, X^{-N} (\log X)^{M_N} \qquad (X \geq X_N), \tag{37}$$

where $\big(\mathrm{Trunc}_{<N} F\big)(X) = \sum_{\nu_t F \leq n < N} p_n(\log X) X^{-n}$ is the actual finite sum obtained by substituting the generators. Equivalently, $f - \mathrm{Trunc}_{<N} F = \mathcal{O}_{X\to+\infty,\ X\in S}\big(X^{-N}(\log X)^{M_N}\big)$.

Lemma 1.44 (Basic properties of $\approx$). *Let $f, g\colon S \to \mathbb{C}$ and $F, G \in \mathscr{PL}_{\mathrm{poly},\mathbb{K}}$.*

1. *If (37) holds for arbitrarily large N, it holds for every N; so it suffices to verify it along a cofinal set of cutoffs.*
2. *$f \approx F$ and $f \approx F'$ imply $F = F'$.*
3. *If $f \approx F$ and $g \approx G$, then $f + g \approx F + G$ and $\lambda f \approx \lambda F$ for $\lambda \in \mathbb{K}$.*
4. *$f \approx 0$ if and only if $f = \mathcal{O}_{X\to+\infty}\big(X^{-N}\big)$ for every N.*

Proof. (1) Let $N' < N$ and suppose (37) holds at N. Then

$$f - \mathrm{Trunc}_{<N'} F = \big(f - \mathrm{Trunc}_{<N} F\big) + \sum_{N' \leq n < N} p_n(\log X) X^{-n},$$

a finite sum; the first term is $\mathcal{O}_{X\to+\infty}\big(X^{-N}(\log X)^{M_N}\big) \subseteq \mathcal{O}_{X\to+\infty}\Big(X^{-N'}(\log X)^{M_N}\Big)$ since $N \geq N'$, and each term of the sum is $\mathcal{O}_{X\to+\infty}\Big(X^{-N'}(\log X)^{\deg_{\mathrm{L}} p_n}\Big)$.

(2) Suppose $D = F - F' \neq 0$, with $n_1 = \nu_t D$ and leading block p of degree d and leading coefficient $c \neq 0$. Take $N = n_1 + 1$ and subtract the two instances of (37):

$$\big(\mathrm{Trunc}_{<n_1+1} D\big)(X) = p(\log X) X^{-n_1} = \mathcal{O}_{X\to+\infty}\big(X^{-n_1-1}(\log X)^{M}\big)$$

for some M. But $|p(\log X)| \sim |c|(\log X)^d$, so the left side divided by $X^{-n_1-1}(\log X)^M$ is $\sim |c| X (\log X)^{d-M} \to +\infty$, a contradiction.

(3) Immediate from $\mathrm{Trunc}_{<N} F + G = \mathrm{Trunc}_{<N} F + \mathrm{Trunc}_{<N} G$ and the triangle inequality.

(4) If $f \approx 0$ then $\mathrm{Trunc}_{<N} 0 = 0$ and $f = \mathcal{O}_{X\to+\infty}\big(X^{-N}(\log X)^{M_N}\big)$ for every N; since $(\log X)^{M_N} = o_{X\to+\infty}(X)$, this gives $f = \mathcal{O}_{X\to+\infty}\big(X^{-N+1}\big)$ for every N, hence $f = \mathcal{O}_{X\to+\infty}\big(X^{-N}\big)$ for every N. The converse is trivial. □

Theorem 1.45 (Analytic product rule). *Let $\mathbb{K} \subseteq \mathbb{C}$, $S = [X_0, +\infty)$ with the real logarithm, and let $f, g\colon S \to \mathbb{C}$ with $f \approx F$ and $g \approx G$ for $F, G \in \mathscr{PL}_{\mathrm{poly},\mathbb{K}}$. Then*

$$fg \approx FG,$$

where FG is the formal Cauchy product (5). Consequently the set $\mathcal{A}(S) = \{f : \exists F \in \mathscr{PL}_{\mathrm{poly},\mathbb{K}},\ f \approx F\}$ is a $\mathbb{K}$-algebra, the map $f \mapsto F$ is a well-defined $\mathbb{K}$-algebra homomorphism $\mathcal{A}(S) \to \mathscr{PL}_{\mathrm{poly},\mathbb{K}}$, and its kernel is the ideal of functions that are $\mathcal{O}_{X\to+\infty}\big(X^{-N}\big)$ for every N.

Proof. First dispose of the degenerate case. Any g with $g \approx G$ satisfies $g = \mathcal{O}_{X\to+\infty}(X^c(\log X)^m)$ for some $c \in \mathbb{Z}$ and $m \in \mathbb{N}_0$: take $N = 1$ in (37), note that $\mathrm{Trunc}_{<1} G$ is a finite sum of monomials $t^n L^j$, and set $c = \max(-\nu_t G, -1)$ (with $c = -1$ if $G = 0$). Hence, if $F = 0$, then Theorem 1.44(4) gives $f = \mathcal{O}_{X\to+\infty}\big(X^{-P}\big)$ for every P, so $fg = \mathcal{O}_{X\to+\infty}\big(X^{-P+c}(\log X)^m\big)$ for every P, whence $fg \approx 0 = FG$ by Theorem 1.44(4) again; the case $G = 0$ is symmetric. Assume from now on $F, G \neq 0$ and set $a = \nu_t F$, $b = \nu_t G$.

Fix $N \in \mathbb{Z}$ and put

$$P = \max(N - b,\ a), \qquad Q = \max(N - a,\ b).$$

Write $f_P = (\mathrm{Trunc}_{<P} F)(X)$, $r = f - f_P$, $g_Q = (\mathrm{Trunc}_{<Q} G)(X)$, $s = g - g_Q$. By hypothesis $r = \mathcal{O}_{X\to+\infty}(X^{-P}(\log X)^{m_1})$ and $s = \mathcal{O}_{X\to+\infty}(X^{-Q}(\log X)^{m_2})$ for suitable exponents. Since $\mathrm{Trunc}_{<P} F$ is a finite sum of blocks with outer exponents $\geq a$,

$$f_P = \mathcal{O}_{X\to+\infty}(X^{-a}(\log X)^{m_3}), \qquad g_Q = \mathcal{O}_{X\to+\infty}\left(X^{-b}(\log X)^{m_4}\right)$$

with $m_3 = \max_{a\leq n<P} \deg_{\mathrm{L}} p_n$ and similarly for m_4 (the estimates being vacuously true if the sums are empty). Therefore

$$fg - f_P g_Q = f_P s + r g_Q + rs$$

is, term by term,

$$\mathcal{O}_{X\to+\infty}\left(X^{-(a+Q)}(\log X)^{m_3+m_2}\right)+\mathcal{O}_{X\to+\infty}\left(X^{-(P+b)}(\log X)^{m_1+m_4}\right)+\mathcal{O}_{X\to+\infty}\left(X^{-(P+Q)}(\log X)^{m_1+m_2}\right).$$

Now $a+Q \geq a+(N-a) = N$, $P+b \geq (N-b)+b = N$, and $P+Q \geq a+(N-a) = N$; hence

$$fg - f_P g_Q = \mathcal{O}_{X\to+\infty}\left(X^{-N}(\log X)^{M'}\right) \tag{38}$$

for some M'.

Next, $f_P g_Q$ is the value at X of the finite polynomial–logarithmic expression $(\mathrm{Trunc}_{<P} F)(\mathrm{Trunc}_{<Q} G) \in \mathscr{P}\mathscr{L}_{\mathrm{poly},\mathbb{K}}$, because substitution of the generators $t = X^{-1}$, $L = \log X$ into a finite sum of monomials is a ring homomorphism into functions on S. Since $P \geq N - b$ and $Q \geq N - a$, Theorem 1.13 gives

$$\mathrm{Trunc}_{<N}(\mathrm{Trunc}_{<P} F)(\mathrm{Trunc}_{<Q} G) = \mathrm{Trunc}_{<N} FG.$$

Hence $(\mathrm{Trunc}_{<P} F)(\mathrm{Trunc}_{<Q} G) - \mathrm{Trunc}_{<N} FG$ is a *finite* sum of monomials $c\,t^n L^j$ with $n \geq N$, so its value at X is $\mathcal{O}_{X\to+\infty}\left(X^{-N}(\log X)^{M''}\right)$ with M'' the largest j occurring. Combining with (38),

$$fg - (\mathrm{Trunc}_{<N} FG)(X) = \mathcal{O}_{X\to+\infty}\left(X^{-N}(\log X)^{\max(M',M'')}\right),$$

which is (37) for fg and FG. As N was arbitrary, $fg \approx FG$.

The algebra statements now follow: $\mathcal{A}(S)$ is closed under $+$, scalars and $\cdot$ by Theorem 1.44(3) and the above; the assignment $f \mapsto F$ is well defined by Theorem 1.44(2) and is a homomorphism by the same two statements; its kernel is described by Theorem 1.44(4). ☐

Remark 1.46 (Which hypotheses are actually used, and the sectorial version). The proof uses only three things: that f and g have remainders bounded by X^{-N} times *some* power of $\log X$; that finitely many blocks are involved at each stage; and the formal finite-determination theorem Theorem 1.12, which is what allows the two truncation cutoffs P and Q to be chosen independently of the shape of F and G. No convergence and no uniformity in L is used or obtained.

The same proof works verbatim on an unbounded set $S \subseteq \{z : |\arg z| \leq \theta\}$ with $0 < \theta < \pi$ and the principal logarithm, with X^{-N} replaced by $|X|^{-N}$ and $(\log X)^M$ by $(\log|X|)^M$; the only additional input is $|\log X| \leq \log|X| + \theta \leq 2\log|X|$ for $|X| \geq \mathrm{e}^{\theta}$, so that a bound in $|\log X|$ and a bound in $\log|X|$ differ only by a constant factor. Uniformity claims on a sector must be stated for a *closed* subsector; the branch of log is part of the statement and cannot be recovered from the formal data.

A common trap

Three cautions, each corresponding to an overstatement found in the sources.

1. *A formal identity is not an analytic theorem.* Theorem 1.45 presupposes that f and g *already have* expansions; it does not produce one. The homomorphism $f \mapsto F$ has a large kernel — $\mathrm{e}^{-X} \approx 0$ — so a formal computation determines an actual function at best up to a flat error.

2. *The formal $\mathcal{O}_t^{\mathrm{form}}(t^N)$ carries no uniformity in L.* Asserting $F = G + \mathcal{O}_t^{\mathrm{form}}(t^N)$ says only that the blocks agree below outer order N. In particular the constant C_N and the exponent M_N of (37) are not supplied by any formal statement and must be produced separately.

3. *The three $\mathcal{O}$-roles are distinct.* The formal tail $\mathcal{O}_t^{\mathrm{form}}(t^N)$, the analytic estimate $\mathcal{O}_{X\to+\infty}\big(X^{-N}(\log X)^M\big)$ of (37), and the complexity bound $\mathcal{O}\big((N-a-b)^2 d^2\big)$ of Theorem 1.16 are three different assertions and none follows from the others. A bare $\mathcal{O}$ for an asymptotic claim, with the regime left to context, is never acceptable.

Section summary

Multiplication is Cauchy convolution in $t = X^{-1}$ with exact coefficient arithmetic in $L = \log X$. Every output block below a cutoff is a finite bilinear expression in finitely many input blocks, with the exact range $a \le n \le N-b$, $b \le m \le N-a$; the resulting precision rule is $\mathrm{Trunc}_{<\min(P+b,\,Q+a)} FG$ and it cannot be improved. All four ambient classes are closed and are integral domains, the leading monomial and leading scalar are multiplicative, and the rank-two valuation $v(F) = (\nu_t F, -\deg_{\mathrm{L}} p_{\nu_t F})$ is additive. The log-degree profile obeys the max-plus bound $d_{FG} \le d_F * d_G$, with equality exactly when the maximizing pairs do not cancel; affine profiles are stable under products, powers, and binomial powers, but not under t-adic limits. Binomial powers $(1+U)^\sigma$ are defined for every $\sigma \in \mathbb{K}$ by a summable family, are finitely determined at every order, and satisfy the exponent law. Under printed remainder hypotheses on a stated domain and branch, the formal product computes the asymptotic expansion of the analytic product — and nothing more.